Technical review

Item #2 · edited-gimp.jpg · Run ID: 20261113T232757Z-cbb6f3

Path	`R:\evidence\images\edited-gimp.jpg`
Size (bytes)	105483
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.jpg
Detected type (signature)	JPEG image
Status	Needs attention
Case	RV-SAMPLE-001
Evidence ID	E0012
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`1a37e10d99d05c6ff552cc2ea29b218b133e1ee319b5b477d5ca64989ddff3f5`
md5	`636327633cd387f9e4f80193cd87112e`

Findings

[Medium] Saved by editing or processing software (EXIF_SOFTWARE_EDITOR)

What it shows	The metadata names GIMP 2.10.36 (EXIF Software, XMP CreatorTool, JPEG comment, XMP history).
What it does not prove	Editors are used for legitimate tasks (rotating, resizing, converting); it shows which program saved the file, not what was changed.

[Medium] Edit history in XMP (XMP_EDIT_HISTORY)

What it shows	The XMP history lists 2 event(s); the last by Gimp 2.10 (Linux).
What it does not prove	The history records software actions (saved, converted, cropped…) as written by the software; it can be incomplete or edited and does not show what the picture looked like before.

[Medium] Metadata dates do not agree (METADATA_DATES_INCONSISTENT)

What it shows	EXIF DateTime 2026:05:16 18:02:10 is 2.3 days after DateTimeOriginal 2026:05:14 10:21:33 and a Software tag is present.
What it does not prove	Clocks, time zones and software that rewrites dates all cause this; it does not show when the picture was taken or changed.

[Informational] Standard (IJG) compression tables (QT_STANDARD_IJG)

What it shows	The quantization tables are the IJG standard tables at quality 85.
What it does not prove	Very many programs and devices use these tables; they identify a family of encoders, not one program or camera.

[Informational] Compression tables not in the database (QT_UNKNOWN)

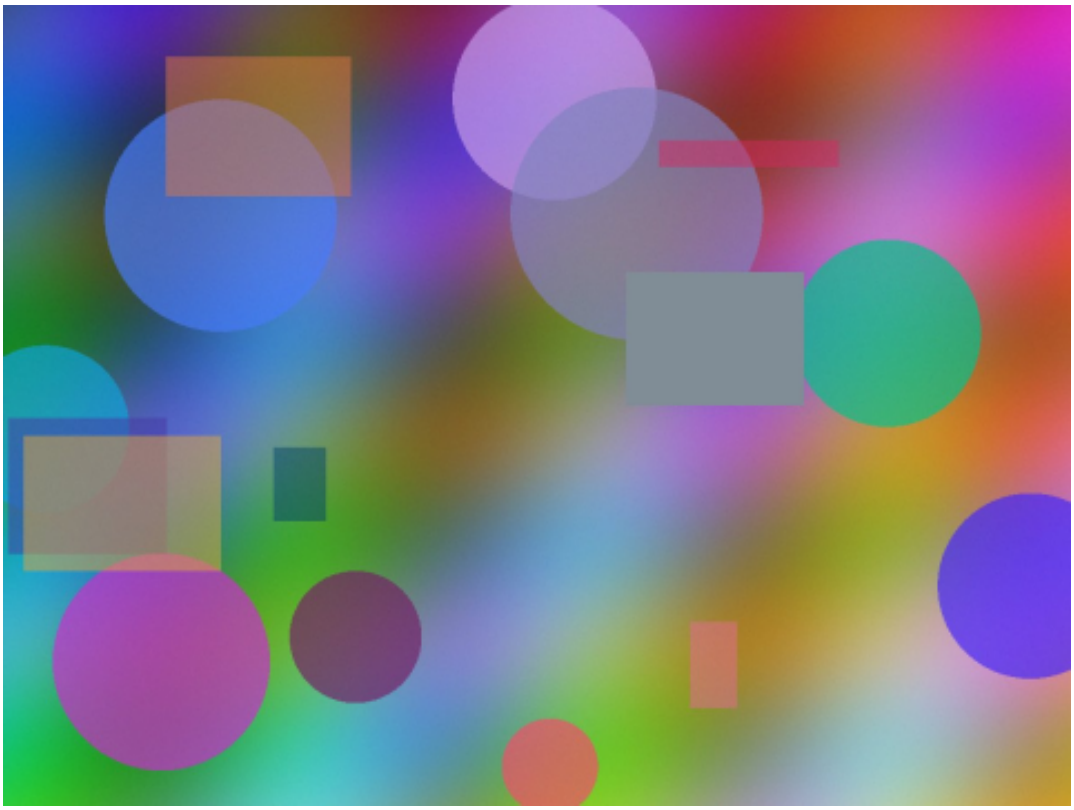
What it shows	No entry of the fingerprint databases matches these tables (SHA-256 43920728cfe021cebc3e0bf197ec4338c16eec09bf89e4b7a5cac6f67651ebfa).
What it does not prove	The database covers few devices and apps; an unknown table says nothing about authenticity. Add known originals with `fingerprints learn`.

Image analysis

Status	Completed
Format	JPEG
Dimensions (pixels)	1200 × 900

Original image (decoded preview)

Lossless PNG of the decoded pixels, exactly as the analyses saw them (stored orientation; EXIF rotation not applied). The maps below have the same size and can be laid over it.



maps/original.png (1200x900) — reduced copy for print; the full-resolution file is in the item folder

Metadata summary

Key fields only; metadata.json in the item folder holds every tag read. Metadata is written by the device or program that saved the file and can be edited or forged.

Tags read	39
Tags per group	COM 1, Exif 13, IFD0 8, IFD1 6, XMP 11
GPS tags	absent
MakerNote	present
XMP packet	present
Embedded thumbnail	present
ICC profile (SHA-256)	—

Group	Field	Value
IFD0	Make	Canon
IFD0	Model	Canon EOS R6

Group	Field	Value
IFD0	Software	GIMP 2.10.36
IFD0	DateTime	2026:05:16 18:02:10
Exif	DateTimeOriginal	2026:05:14 10:21:33
Exif	DateTimeDigitized	2026:05:14 10:21:33
Exif	OffsetTime	+02:00
Exif	OffsetTimeOriginal	+02:00
Exif	PixelXDimension	1200
Exif	PixelYDimension	900
IFD0	Orientation	1 (row 0 at top and column 0 at left)
XMP	xmp:CreatorTool	GIMP 2.10
XMP	xmp:ModifyDate	2026-05-16T18:02:10+02:00
XMP	xmp:MetadataDate	2026-05-16T18:02:10+02:00
COM	Comment	Created with GIMP

XMP edit history (xmpMM:History)

Action	When	Software	Changed
saved	2026-05-16T17:40:02+02:00	Gimp 2.10 (Linux)	/
saved	2026-05-16T18:02:10+02:00	Gimp 2.10 (Linux)	/

Structure summary

Top-level segments in file order (JPEG markers, PNG chunks, RIFF chunks or HEIF boxes); structure.json holds the full tree with byte offsets.

Top-level segments	15
frame (SOF)	SOF0
chroma subsampling	4:2:0
components	3
restart interval	0
quantization tables	2
Huffman tables	4
bytes after EOI	0
comments (COM)	1

Marker / chunk / box sequence:

```
SOI · APP0/JFIF · APP1/Exif · APP1/XMP · COM · DQT ×2 · SOF0 · DHT ×4 · SOS · EOI
```

Quantization tables

Values in natural (row-major) order, as stored in the DQT segments. The IJG comparison names a standard libjpeg quality only on an exact match; otherwise the nearest quality and its distance (sum of absolute differences) are given, which is not a quality percentage.

IJG comparison	IJG-standard q=85
Database match	no entry in the loaded databases
Table hash (SHA-256)	`43920728cfe021cebc3e0bf197ec4338c16eec09bf89e4b7a5cac6f67651ebfa`

Table 0 · 8-bit · Used by components: 1

```
5   3   3   5   7  12  15  18
4   4   4   6   8  17  18  17
```

```
4   4   5   7  12  17  21  17
4   5   7   9  15  26  24  19
5   7  11  17  20  33  31  23
7  11  17  19  24  31  34  28
15  19  23  26  31  36  36  30
22  28  29  29  34  30  31  30
```

Table 1 · 8-bit · Used by components: 2, 3

```
5   5   7  14  30  30  30  30
5   6   8  20  30  30  30  30
7   8  17  30  30  30  30  30
14  20  30  30  30  30  30  30
30  30  30  30  30  30  30  30
30  30  30  30  30  30  30  30
30  30  30  30  30  30  30  30
30  30  30  30  30  30  30  30
```

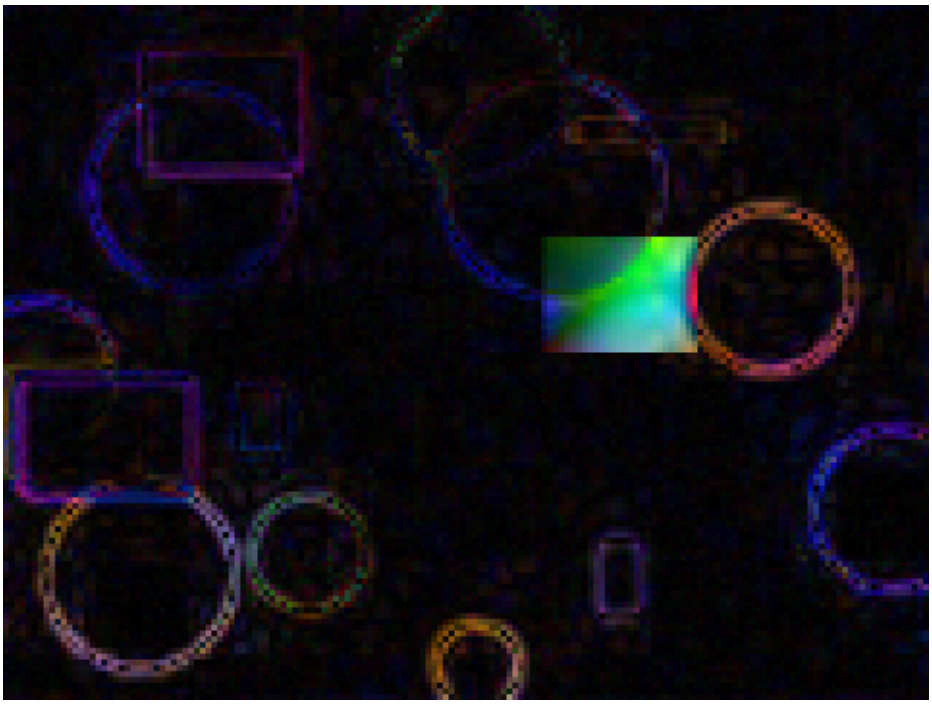
Fingerprint databases

Role	Database	Version	Entries	Status	SHA-256
starter	rakvor-starter	0	0	loaded	e0b0c87e4b79e5c5402e920d343436056c25bb959a5bb757a7d0519b95ab4be8
local	—	—	—	absent	—

Analysis maps

Visual aid, requires interpretation: these maps have no score and the tool draws no conclusion from them. Bright areas are not proof of a change; authentic edges, textures and noise produce them too.

Thumbnail vs main image difference



maps/thumbnail-diff.png — reduced copy for print; the full-resolution file is in the item folder

compared_area	160x120 at 0,0 (black bars trimmed)
gain	4
largest_block_difference	33.1
mean_difference	3.5

resize	main image area-averaged to the thumbnail size
thumbnail_size	160x120
upscale	4

Copy-move shift vectors

No copy-move candidate was found (scaled or rotated copies are not detected).

Extracted files

structure.json (structure, 11358 B)	md5: 47b0cd5763e460c0a4fa48bdceeaf932 sha256: 74bb3029b6ddf4b6fee3aee320cb722c4f3f99c79be20d813006525fa4e12ef9
metadata.json (metadata, 6405 B)	md5: 455bd7695741478667ed2cf21623f740 sha256: 6d9a9d70212eb2428a487968e14dba0902f63b07590114e2746e4ca42970ad28
maps/original.png (original_preview, 1076768 B)	md5: 07cc9ec0c4cc1933390357eeabc62b92 sha256: 8e9e44761c4d9c9b1eb1d4811de96affdf0257986d9d9251bbae7593bf8ce038
maps/thumbnail-diff.png (map, 70868 B)	md5: 61d8546f750f09cb20df0cb97fd4cbc9 sha256: d9b738f183865a3326a9de0cc997f0a866f27c713fcaf23c7f97e4ede642e334

Method

Each file was opened read-only and hashed in a single pass with the selected algorithms, and its real type was identified from its signature. JPEG, HEIC, PNG and WebP images were then re-read in an isolated process and the bytes read were hashed again and compared with the first pass. The image analysis read the file structure and metadata, compared the JPEG quantization tables with the IJG standard and the fingerprint databases, looked for double JPEG compression, wrote the selected visual maps (error level, noise, JPEG ghosts), searched for copied-and-moved regions, compared the embedded thumbnail with the main image and, when reference images were given, compared the image with them. HEIC pixels are not decoded (structure and metadata only). Every action and every file written was recorded in a hash-chained audit log.

Audit trail

Audit entries	#14 – #22
Log folder	`R:\case\runs\rv-imgauth\20261113T232757Z-cbb6f3\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- An image without findings is not proven authentic: careful edits can leave none of the traces these checks look for, and it is impossible to prove a negative (SWGDE 18-I-001).
- Error level, noise and ghost maps are visual aids for a trained examiner; the tool gives them no score and draws no conclusion from them.
- Double JPEG compression shows the image was saved more than once; ordinary sharing and exporting do this.
- The compression-table database covers few devices and apps (built from Rakvor's own captures plus local learning); an absent or matching entry is weak evidence on its own.
- Copy-move detection compares blocks at the same scale and orientation on a downscaled image: scaled, rotated or small copies are not found.
- Error rates of these checks have not yet been measured on a documented validation corpus; the findings are signals to be weighed by the examiner.

- Modification times come from the file system of the examined copy and may differ from the original source.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.
- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.